

Foundational Skills Program

Course Title: Software Testing



Date: April 19, 2026

Md. Golam Mostofa

AI/ML-QA Engineer, Bizzntek Ltd

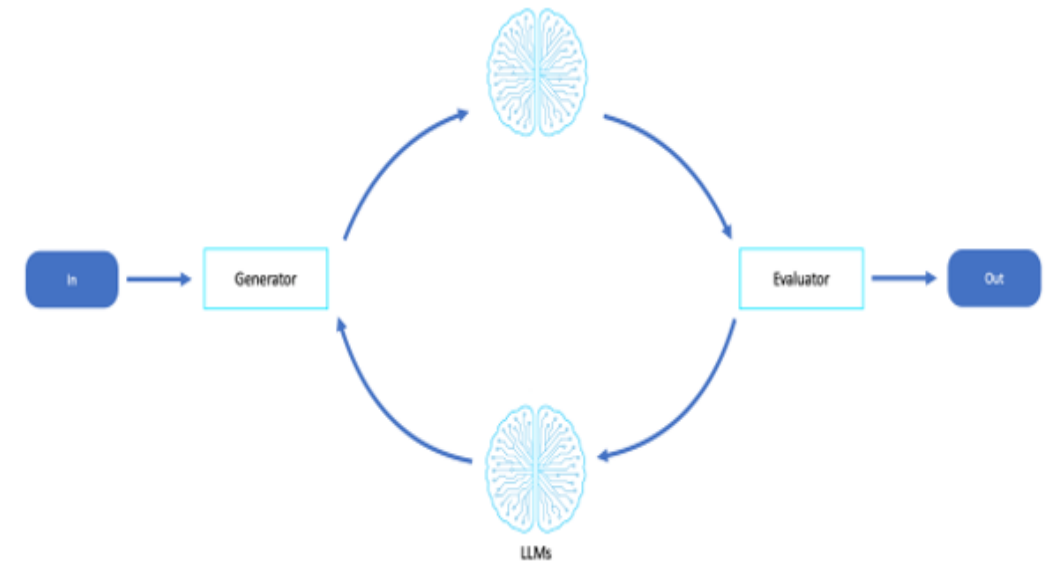


Outline

1. Evaluators and reflect-refine loops
2. Introducing Mistral LLM
3. Mistral API using
4. Evaluation Setup
5. Environment Setup
6. AI-Driven Automated Testing of EDGE Chatbot

Evaluators and reflect-refine loops

- Evaluator-optimizer workflows use two LLM stages: one generates the response, and another evaluates its quality.
- The evaluator checks whether the generated output meets predefined success criteria.
- These workflows are effective for tasks that require refinement and multiple iterations.
- This approach enables autonomous quality assurance for LLM-based systems and chatbots.



Source: [Agentic AI patterns and workflows on AWS - AWS Prescriptive Guidance](#)

Common use cases

- Red-team agents compared to blue-team agents
- Agents that generate, evaluate, and revise code or plans
- Quality assurance, hallucination detection, and style enforcement

Introducing Mistral LLM



What is Mistral AI?

- [Mistral AI](#) is a France-based AI company focused on building open and efficient Large Language Models (LLMs).

Popular Mistral models

- **Mistral 7B:** lightweight and efficient open-weight LLM
- **Mixtral 8x7B:** Mixture-of-Experts (MoE) model for improved scalability
- **Mistral Large:** flagship model for advanced reasoning and enterprise tasks
- **Codestral:** optimized for coding and software development
- **Le Chat:** conversational AI assistant similar to ChatGPT

Mistral emphasizes

- Open-source and open-weight AI models
- Multilingual support
- Efficient deployment on smaller hardware
- Privacy-focused and on-premise AI solutions

Mistral API using

`</> Python`

```
from mistralai import Mistral

api_key = "YOUR_API_KEY"

client = Mistral(api_key=api_key)

response = client.chat.complete(
    model="mistral-large-latest",
    messages=[
        {
            "role": "user",
            "content": "Explain Retrieval-Augmented Generation"
        }
    ]
)

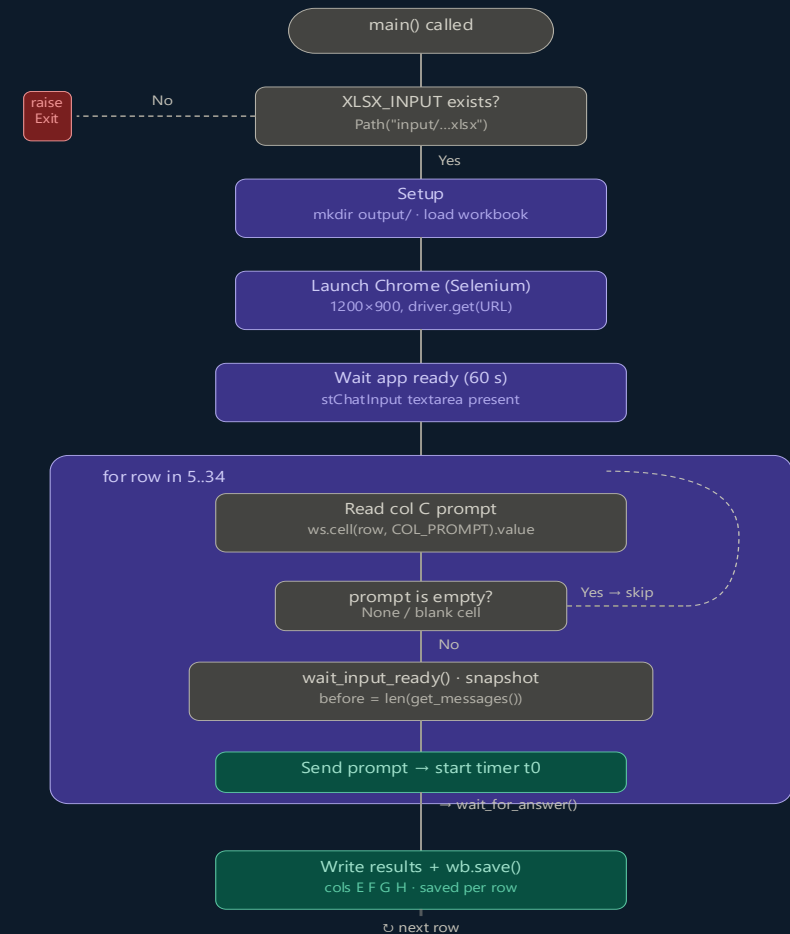
print(response.choices[0].message.content)
```

Steps to Use the Mistral API

- Create an account on [Mistral AI Platform](#)
- Generate an API key from the developer console
- Install the official SDK or use HTTP requests
- Send prompts to supported models such as:
 - Mistral 7B
 - Mixtral 8x7B
 - Mistral Large
 - Codestral

EDEG Chatbot Test automation Architecture

- Start from main function
- Checking test file available or not
- Result folder checking
- Launch app in Chrome browser
- Wait for ready
- Put prompt into the chatbot
- Get response
- Test response by the Mistral LLM
- Same test result



Pre-Requisites

```
root@ae68fead5916:/# ollama list
```

NAME	ID	SIZE	MODIFIED
llama3.2:1b	baf6a787fdff	1.3 GB	29 seconds ago
nomic-embed-text:latest	0a109f422b47	274 MB	About a minute ago

```
root@ae68fead5916:/#
```

Key Features



Ollama install



Chrome-browser



Llama3.2:1b



Nomic-embed-text

Environment Setup For EDEG Chatbot



1 Create Virtual Environment & Active

```
python -m venv venv  
venv\Scripts\activate
```

Download the [EDEG Chatbot](#) first

2 Install Requirements

```
pip install -r requirements.txt
```

Install the Python packages.

3 Run the Chatbot

```
streamlit run app.py
```

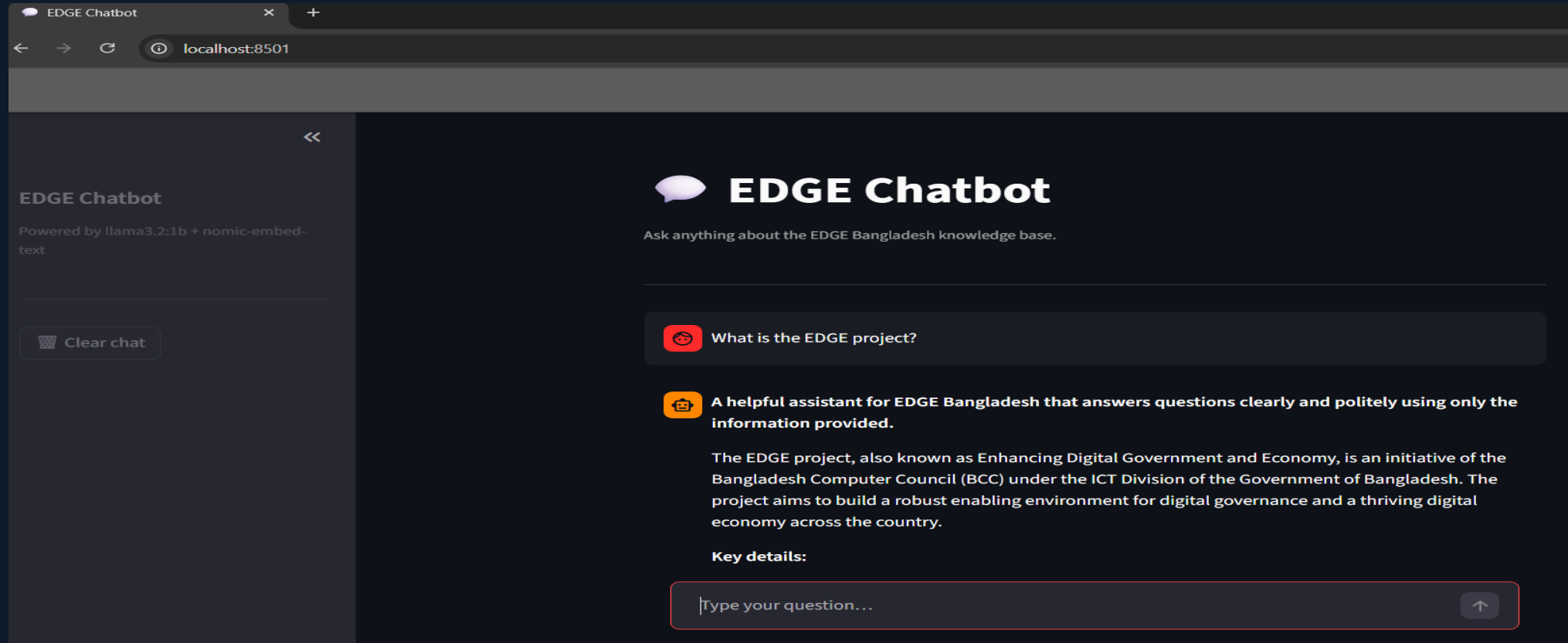
Navigate: <http://localhost:8501/>

4 Test the Chatbot

```
venv\Scripts\activate  
python fill_test_sheet.py
```

Run the test.

Test EDGE Chatbot Automatically



Q&A

Any Questions?



Thanks



MD. GOLAM MOSTOFA



golam.mostofa@bizzntek.com